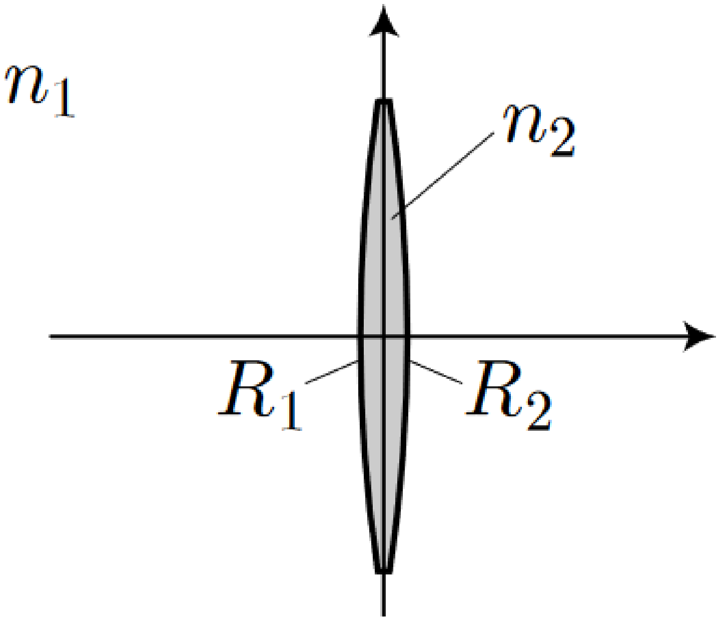
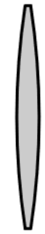
Ph134, Lecture 14

- **Thin lenses:**
  - **Lensmaker's equation**
  - **Converging and diverging lenses**
  - **Thin lens equation**

- **Imaging with systems of lenses:**
  - **Real and virtual images**
  - **Real and virtual objects**
  - **It's all about how the rays are pointing!**

ABCD matrix for thin lens:





$R_1 > 0, R_2 < 0$

$R_1 < 0, R_2 > 0$

$$\mathbf{M} = \begin{bmatrix} 1 & 0 \\ \dfrac{(n_2 - n_1)}{n_1}\left(\dfrac{1}{R_2} - \dfrac{1}{R_1}\right) & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -\dfrac{1}{f} & 1 \end{bmatrix}$$

*f is the **focal length** of the lens.*

$$\frac{1}{f} = \frac{(n_2 - n_1)}{n_1}\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

the lensmaker's equation

What does the focal length mean?

$$\begin{bmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{bmatrix} \begin{bmatrix} y_{in} \\ \theta_{in} \end{bmatrix} = \begin{bmatrix} y_{in} \\ \theta_{in} - \frac{y_{in}}{f} \end{bmatrix}$$

$f > 0$? Lens bends incoming rays TOWARD the optical axis: Converging lens

$f < 0$? Lens bends incoming rays AWAY FROM the optical axis: Diverging lens

Smaller magnitude of f means lens bends rays more.

Converging ($f > 0$) and diverging ($f < 0$) lenses in air:

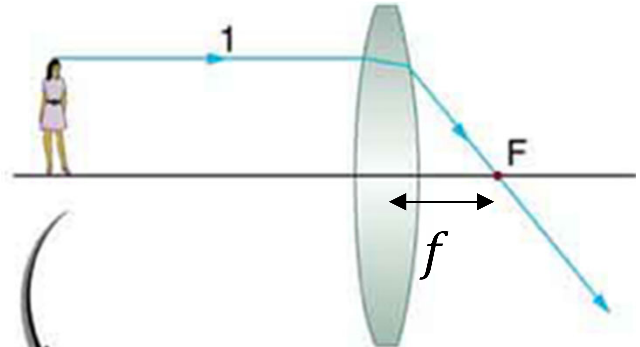
$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Light incident on convex surface: R>0
Light incident on concave surface: R<0

Ray-tracing rules

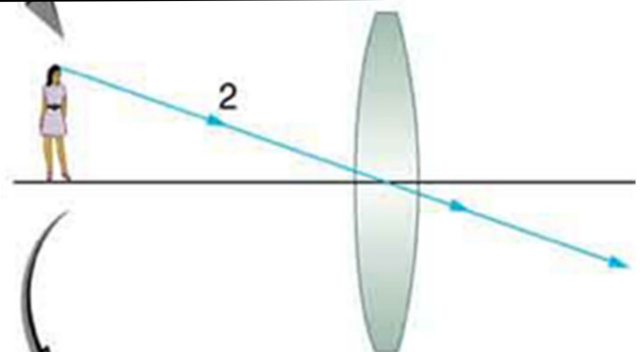
Incoming ray parallel to OA?

Outgoing ray intersects OA at f downstream of lens



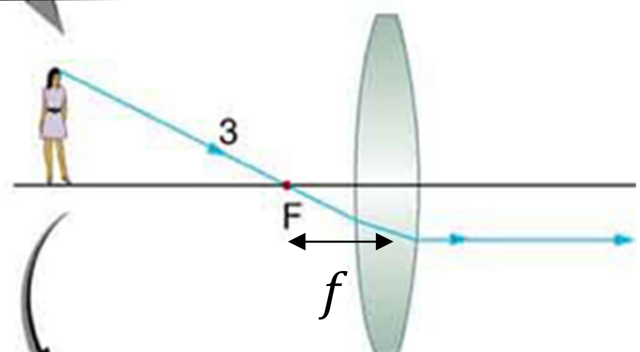
Incoming ray through OA?

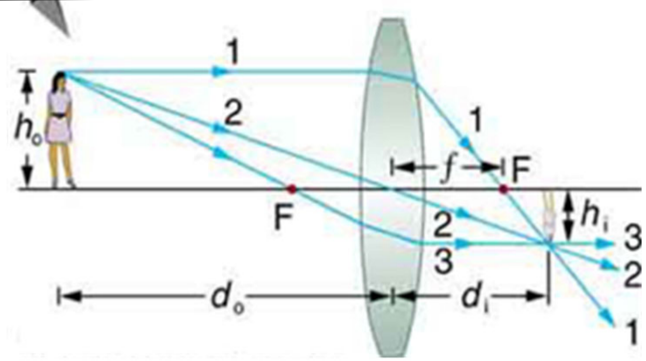
Outgoing ray not deflected.



Incoming ray intersects OA at f upstream of lens?

Outgoing ray parallel to OA



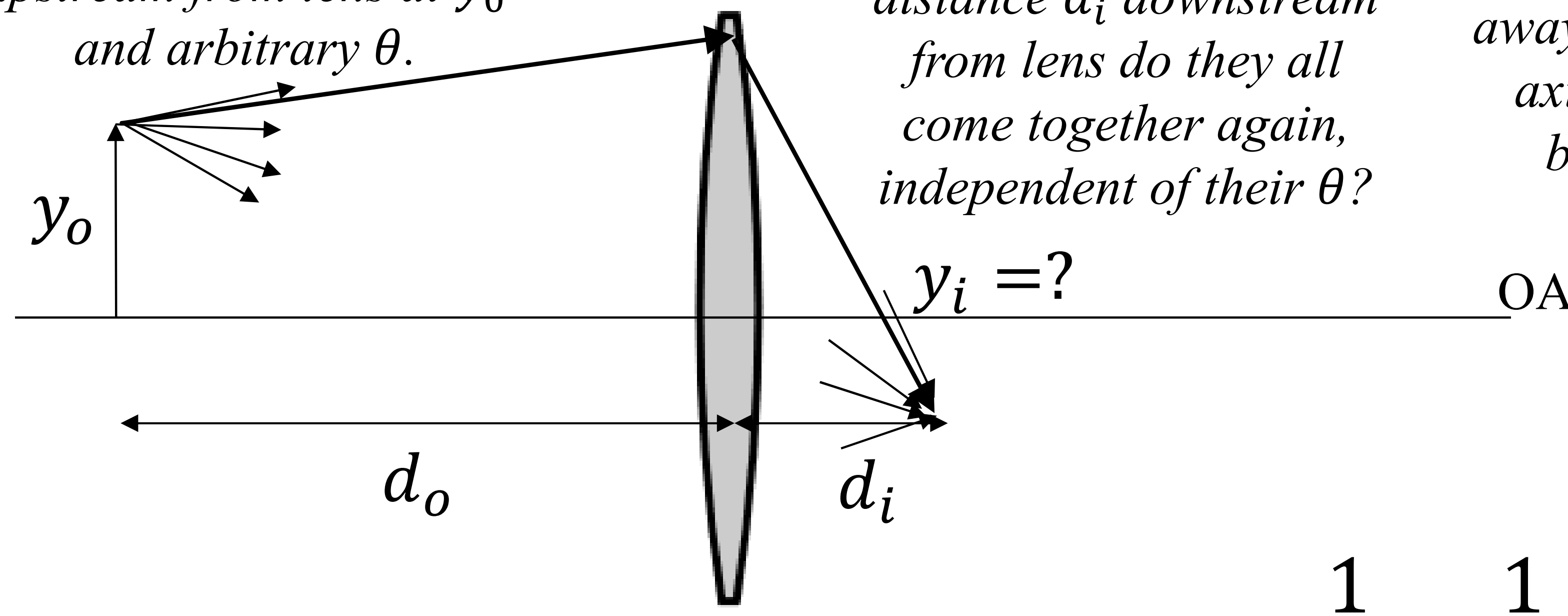


Thin lens equation and magnification:

Consider rays that start longitudinal distance d_0 upstream from lens at y_0 and arbitrary θ.

At what longitudinal distance d_i downstream from lens do they all come together again, independent of their θ?

At what position away from the optical axis do they come back together?



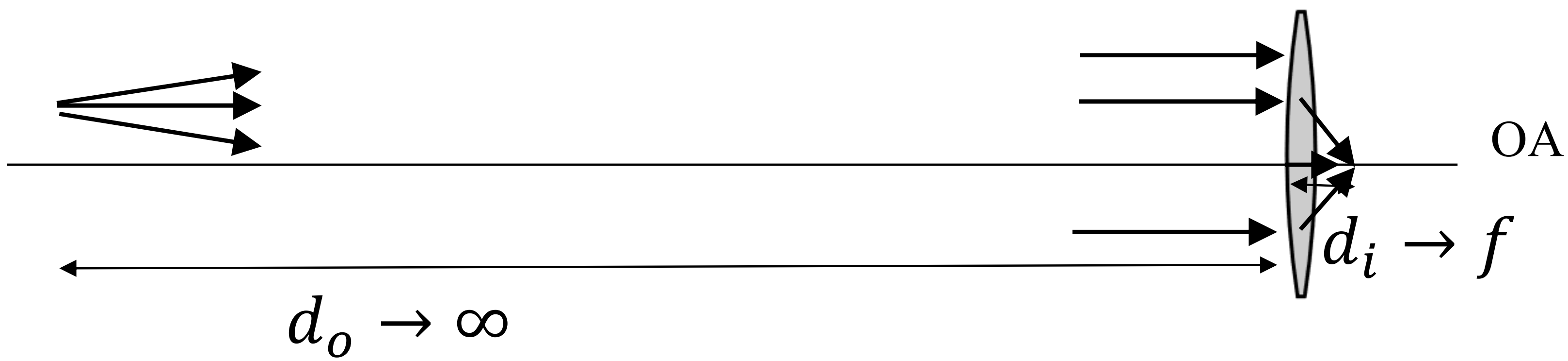
$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

$$Magnification \quad \frac{y_i}{y_o} = -\frac{d_i}{d_o}$$

Is the focal point the place where the lens forms a focused image? Only if the object being imaged is infinitely far away!

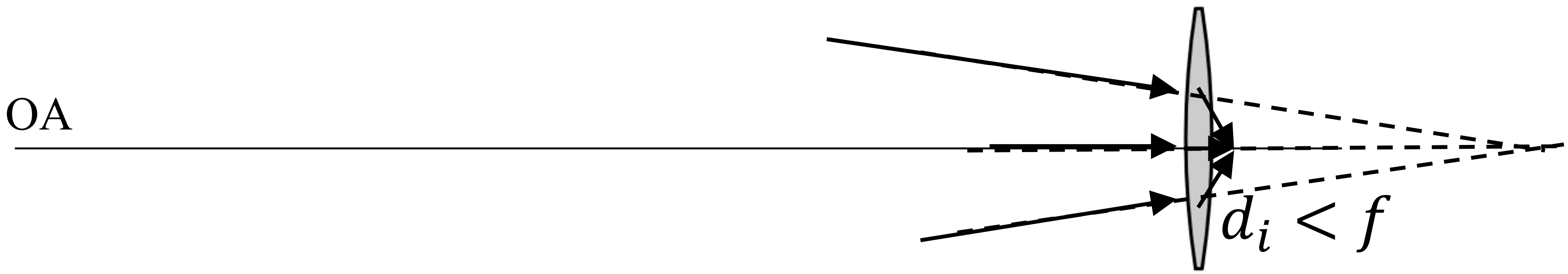
$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

An object at infinite distance will be imaged at f downstream from the lens.



Or: an incoming collimated beam (a beam with near-planar wavefronts) will be focused to a spot at f downstream from the lens.

Can we put the object farther away than ∞ ?



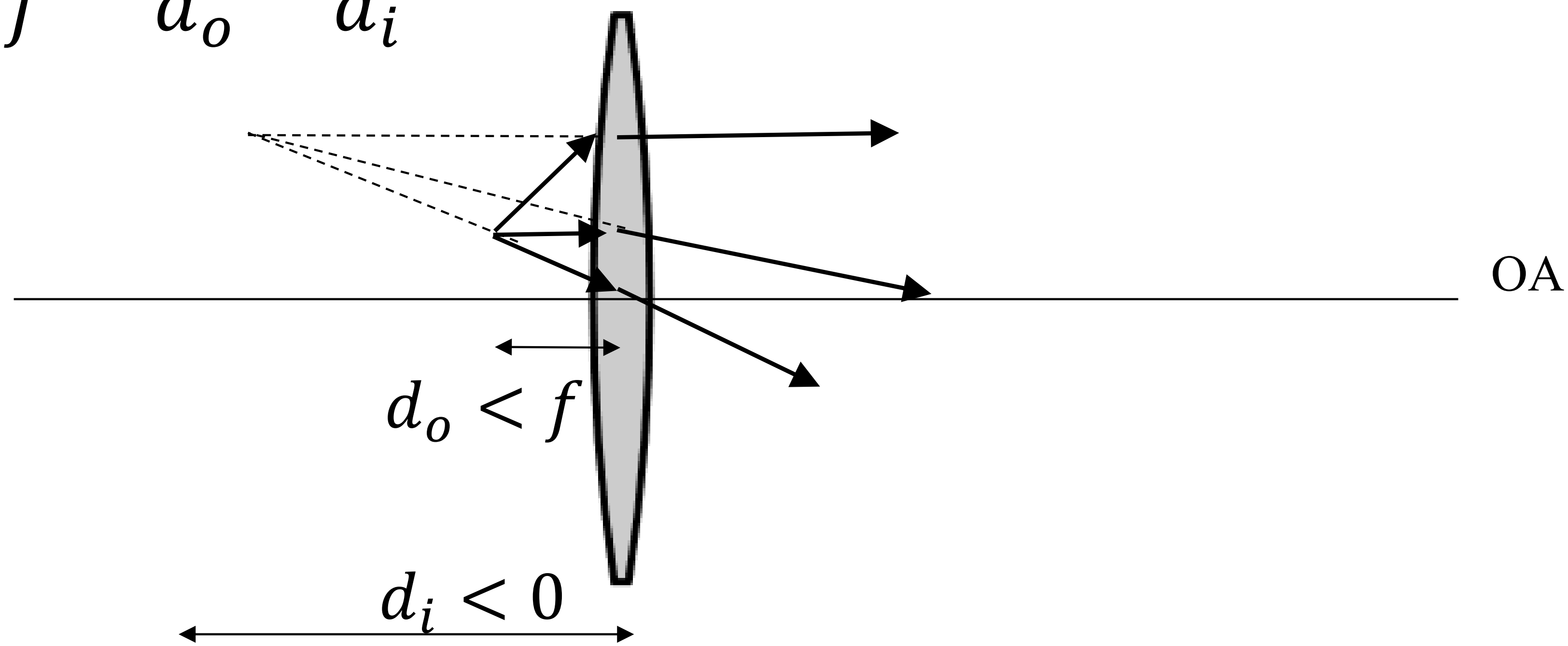
$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

$$d_o < 0$$

Rays incident on lens are already converging – on path as if coming from object at a position DOWNSTREAM of the lens.

We tried huge d_o… what about tiny?

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$



Nothing at the image position, no light propagating "back" to it.
But rays downstream of lens <u>look like</u> they came from that spot.

Real and virtual images:

Real image: ***downstream*** *of lens.* $d_i > 0$

Light rays coming from lens actually pass through image pos'n.

Virtual image: ***upstream*** *of lens.* $d_i < 0$

Light rays coming from lens are <u>angled as if</u> they pass through image pos'n.

Real and virtual objects:

Real object: ***upstream*** *of lens.* $d_o > 0$

Light rays incident on lens actually come directly from each point on the object.

Virtual object: ***downstream*** *of lens.* $d_o < 0$

Light rays incident on lens are <u>angled as if</u> they are converging on (or if time reversed, diverging from) the "object."

The human eye and corrective lenses:

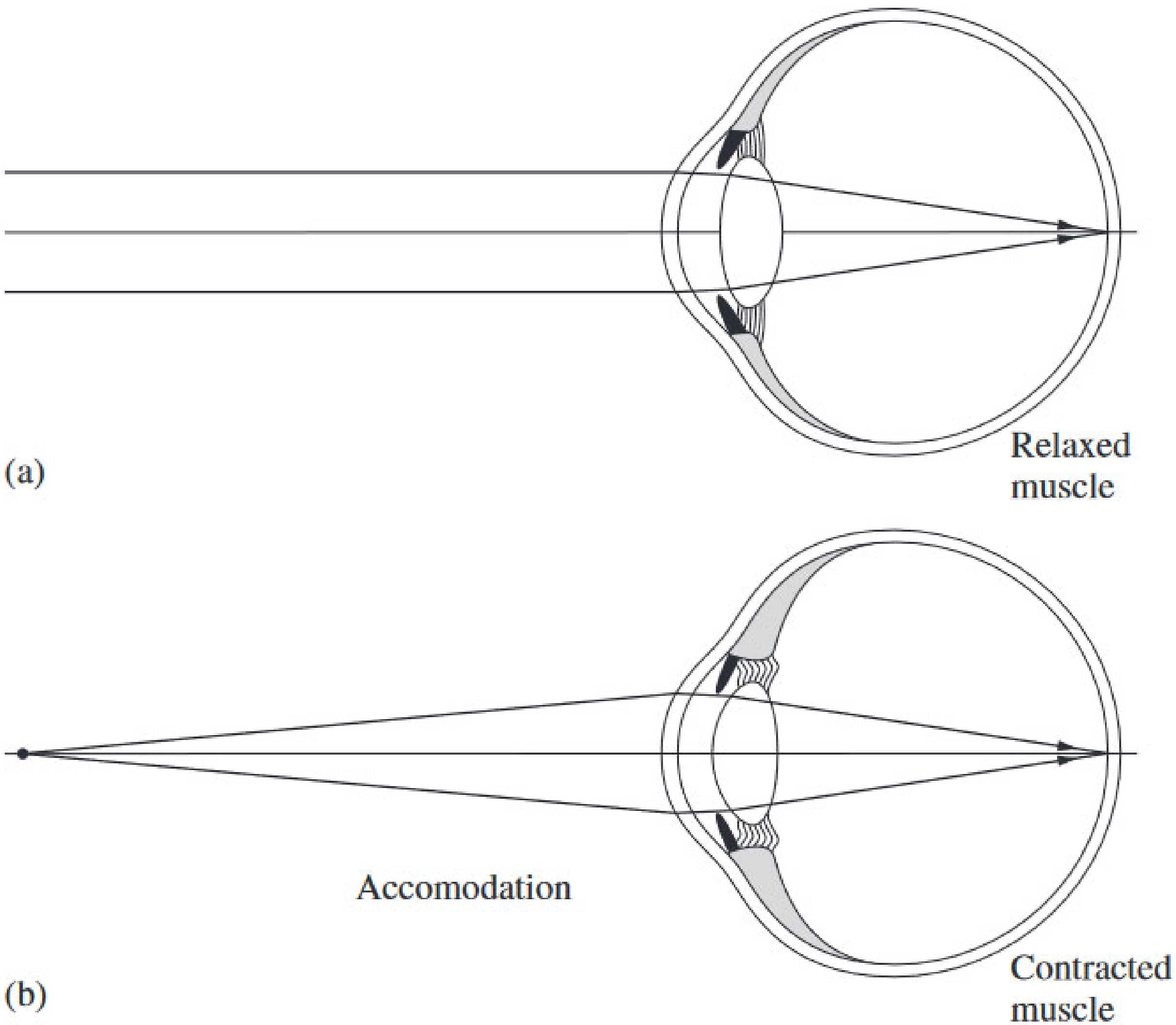
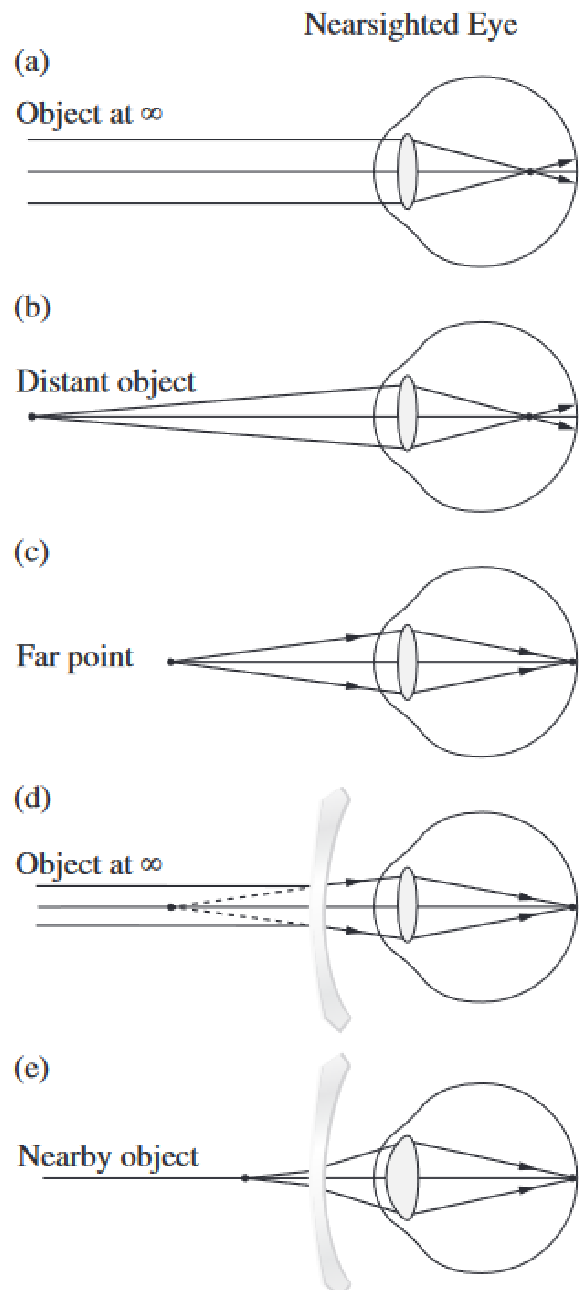


Figure 5.93 Accommodation—changes in the lens configuration.

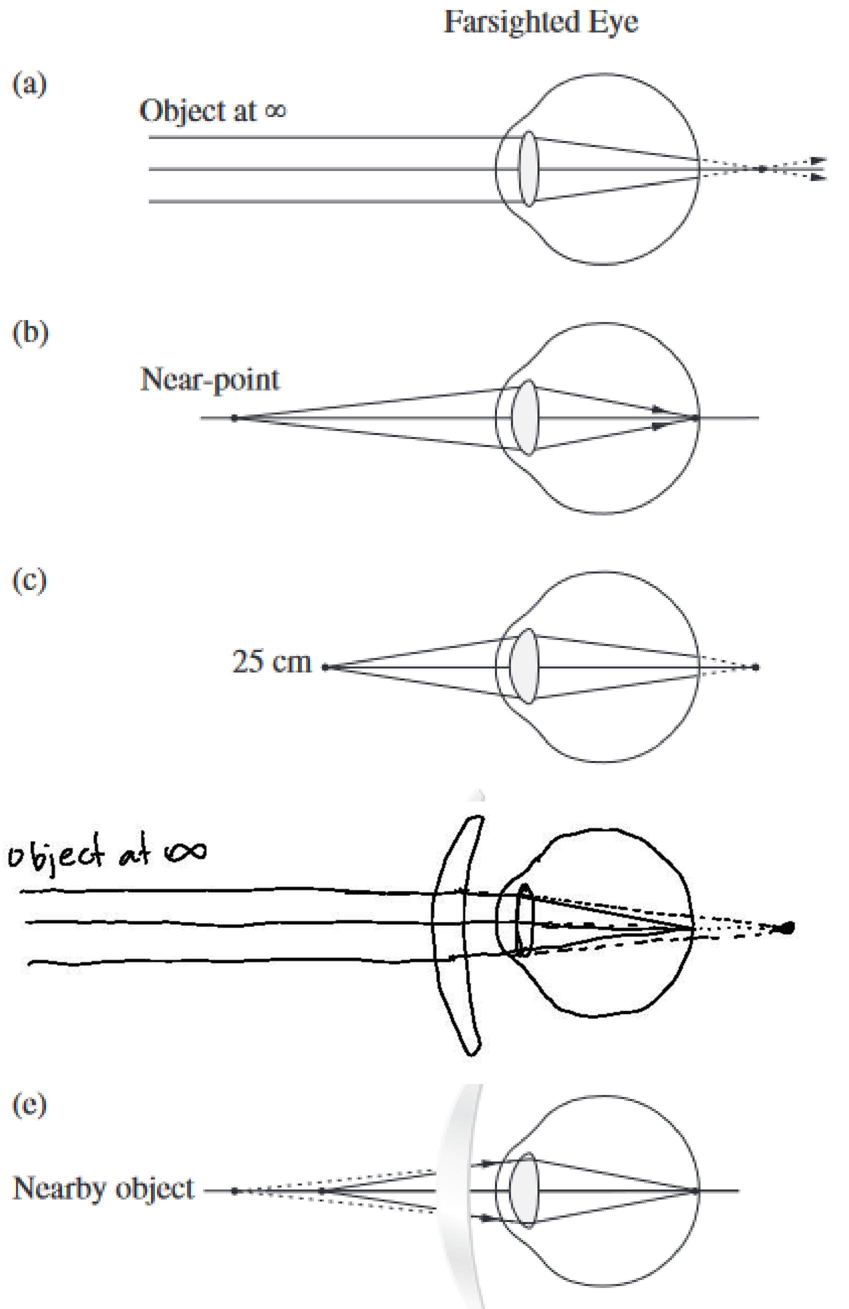
Figures from Hecht, Optics

The human eye and corrective lenses:



Figures from Hecht, Optics

The human eye and corrective lenses:



Figures adapted from Hecht, Optics